eedi-mining-misconceptions-in-mathematics

Private 9th (Public 7th) Place Solution

Rank	9
Team	Study Hard
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Topic ID	551420
Version	Original

Source: https://www.kaggle.com/competitions/eedi-mining-misconceptions-in-mathematics/discussion/551420
Retrieved with Kaggle CLI on 2026-10-21.

I must show my appreciation with respect to the eedi organization and kaggle team, for hosting such a stunning competition. This is my first participation of a kaggle competition, and I am very happy to take the gold medal.

Overview

1. Synthetic data generation (GPT-4o)
2. Rationale generation (Qwen 2.5 32B Instruct AWQ with vllm)
3. Retriever (Qwen 2.5 14B Instruct)
4. Reranker (Qwen 2.5 14B Instruct and Qwen 2.5 32B Instruct with vllm)

Synthetic data generation

We generated SubjectName, ConstructName, QuestionText, Correct Answer, Incorrect Answer by providing misconceptions to GPT-4o.
Next, start filtering:

- For misconceptions in `train.csv`, we filtered generated data with base model (training Qwen 2.5 14B Instruct without synthetic data scored 0.447 on LB).
- For misconceptions not in `train.csv`, do nothing.

Rationale generation

Input the SubjectName, ConstructName, QuestionText, Correct Answer and Incorrect Answer into Qwen 2.5 32B Instruct AWQ to generate a briefly rationale (up to 50 words).
Prompt:

```
Prompt =
```

Retriever

Following @sayoulala and his discussion https://www.kaggle.com/competitions/eedi-mining-misconceptions-in-mathematics/discussion/543519.

- Upsample original trainset question once for training, where misconceptions occur less than 3.
- Training retriever by contrastive learning and using sfr for hard negative sample mining.

Reranker

- Binary classification reranker (14B), predict whether incorrect answer and misconception are matched. (we didn't use it finally because of unefficiency)
- Causal LM reranker (32B), using lm head take yes token logits as ranking score.Once trained, AWQ quantization is performed.

32B inference referer to https://www.kaggle.com/code/cdeotte/infer-34b-with-vllm.

Model	Public LB	Private LB
Retriever	0.547	0.491
Binary classification reranker	0.606	0.542
Causal LM reranker	0.641	0.571

btw, we started training Reranker only a week before the competition cut-off and not much time for attempt 😝

Supplemental Q&A

> **xbustc · 2024-12-13T05:18:03.683000**
>
> Big con! I wonder whether you tried the cross entropy loss to train reranker just like flagemedding reranker?

> **Okinawa · 2024-12-13T05:28:57.333000**
>
> yes, we modified code and llm prompt adapting for our solution

> **Nicholas Broad · 2024-12-13T05:55:07.443000**
>
> Nice job! Can you share the private lb scores in the table for retriever, binary classification reranker, and CLM reranker?

> **Nikhil Tumbde · 2024-12-13T06:00:58.723000**
>
> Hi, thanks for the solution. Would you be sharing a detailed solution or inference code? I really wanted to understand how to properly use synthetic data.
> I also have the following questions for you:
> What do you mean by "we filtered generated data with base model" ?
> I still couldn't understand how the generated data and rationales were used in modelling, can you please explain?

> **Okinawa · 2024-12-13T07:24:40.170000**
>
> Base model means the model do not use synthetic data, training on single fold trainset scored 447 on LB.
> I'll make the inference code public soon.

> **Okinawa · 2024-12-13T07:40:48.190000**
>
> Thanks @nbroad , I've made my private score public.

> **Nikhil Tumbde · 2024-12-14T14:23:14.490000**
>
> Hi. Can you elaborate on the "filtered" part? I am unable to understand that. Really appreciate your response.

> **Okinawa · 2024-12-14T16:31:52.373000**
>
> Because the synthetic data generated by GPT4o is inaccurate, these data are filtered by the retriever with the highest score on LB. Specifically, we dropped the synthetic data where their misconception not in recall top 5 or 10 from retriever.